Víctor García Carrasco

PhD in Statistics & Operations Research · Simulation & Optimisation · Energy Systems Modelling · Python

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Profile

Physicist with a PhD from UPC (2026, Excellent), specialising in the simulation and optimisation of complex systems. I build hourly simulation, discrete-event and multi-objective optimisation models in Python and publish them as web tools at orlabs.cloud. Applied experience in the electricity sector, the water-energy nexus and epidemiological simulation, with 7 publications. Looking to bring this profile to engineering, energy, data or R&D projects.

Technical skills

- **Programming:** Advanced Python (NumPy, pandas, SciPy, Numba, joblib, DEAP), with vectorised code, memoisation and process parallelisation, automated batch pipelines with Excel and PDF output, web scraping, development and deployment of web applications.
- **Simulation:** discrete-event (GPSS), system dynamics (Insight Maker), SDL, cellular automata, digital twins, model validation.
- **Optimisation:** multi-objective, evolutionary algorithms, exact and heuristic methods for routing and networks (TSP/VRP).
- **Data analysis and predictive models:** multiple regression and autoregressive models, machine learning and deep learning, time series, design of experiments, scenario analysis and visualisation.
- **Energy domain:** Iberian electricity mix, wholesale prices, curtailment and storage.

Experience

Universitat Politècnica de Catalunya (UPC) — Lecturer · 2023–present

Department of Statistics and Operations Research.

- Teach Simulation (BSc in Computer Engineering) and Discrete Network Models (MSc in Statistics and Operations Research).

- Design lab assignments on discrete-event simulation, design of experiments and system dynamics.

UPC — Doctoral researcher, water-energy nexus in Catalonia · 2021–2026

- Predictive model of the water volume stored in the Catalan internal basins.
- Hourly simulation model integrating renewable generation, electricity demand, reservoirs and desalination.
- Multi-objective optimisation of energy-transition scenarios: emissions, cost and water security.
- Opportunistic desalination strategy driven by renewable energy availability.
- Conference paper published by Springer (DSA ISC 2025).
- Thesis graded Excellent.

UPC — Researcher, COVID-19 epidemiological simulation · 2020–2023

- SDL-PAND project (UPC Centre for Development Cooperation): SEIRD models, digital twin and cellular automata to assess containment strategies in Catalonia.
- Co-author of 5 journal articles and 1 IEEE conference paper.

Personal projects

- **OR Labs** — operations research web platform that I designed, built and deployed, available in three languages with student access. Includes discrete-event simulation (PyGPSS-X), vehicle routing optimisation (Opt-Lab), hourly electricity mix analysis (EnerSim) and agent-based macroeconomic simulation (EcoSim).
- **Spanish electricity market** — work used in the preparation of the Greenpeace report Cierre nuclear y transición energética: el caso de Almaraz (nuclear phase-out and the energy transition). Included a wholesale price model driven by gas prices, with battery damping calibrated on CAISO data, applied to nuclear closure scenarios (Almaraz I and II).

Education

Year	Degree	Institution
2026	PhD — *The water-energy nexus in Catalonia facing the energy transition* (Excellent)	UPC

Year	Degree	Institution
2012	Postgraduate course in Quantitative Techniques for Financial Markets	UPC
2008	MSc in Computational and Applied Physics	UPC
2007	Degree in Physics (5-year Licenciatura)	UB

Publications

7 publications: 5 journal articles and 2 conference papers. Selected:

- García, V.; Fonseca, P. Energy Transition and Sustainable Water Management in Catalonia: Towards a Holistic Simulation Model. *Decision Sciences*, Springer, 2026.
- Fonseca, P.; García, J.; García, V. Modeling SARS-CoV-2 true infections in Catalonia through a digital twin. *Advanced Theory and Simulations*, 2023.
- Fonseca, P.; García, J.; García, V.; Pi, X.; Wainer, G. Formal modeling and simulation for SARS-CoV-2 containment scenarios in Catalonia. *Computing in Science & Engineering*, 2022.

Languages

Catalan and Spanish: native (C2) · English: reading B2, speaking and writing B1